EBIN BABU THOMAS

AI Engineer: LLM Agents, RAG & Evaluation Systems

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Mavelikara, Kerala, India · Remote · Open to relocation: Bengaluru / Hyderabad

SUMMARY

AI engineer building LLM agents, search systems and the tests that check them. Independent since March 2026, shipping a local meeting transcriber in public beta and a PDF-to-Word converter on PyPI. Built hybrid keyword and vector search over 433 city council meetings, and a purchase-checking agent that keeps hashed screenshots as evidence. Before that, 3.5 years building LLM, RAG and backend systems for startup clients.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Anthropic API, Google Gemini API, OpenAI API, PydanticAI, LangGraph, LangChain), MCP (FastMCP), Hybrid RAG (BM25 + RRF), Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, SQLite (FTS), MongoDB, Vector DBs (Qdrant, LanceDB), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), Speech (Whisper, NVIDIA Nemotron ASR + diarization), Image & Document Processing (OpenCV, scikit-image, Pillow, RapidOCR, PDFium, PyMuPDF), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, GKE, OAuth2 / JWT, Terraform, Linux, uv

Evaluation engineering & measurement: eval harness design (Inspect, inspect_evals), retrieval benchmarking, agent evaluation sweeps, replaying recorded agent runs, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, predictions written before runs, hash-pinned reproducible artifacts, Langfuse

Familiar: Rust / Tauri 2, Svelte 5, Bytewax, Next.js, TypeScript

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. Services firm building first versions of products, AI systems and data tools for startup clients in the US, Canada, Europe and Australia. Grew from trainee to core developer; owned client backends end to end and scoped incoming projects.

- **FinBot, financial RAG & fine-tuned LLM:** fine-tuned Mistral 7B (QLoRA) and built a streaming news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite live market news; deployed and served with vLLM on a GKE GPU node.
- **Natural-language property search:** built a plain-English-to-SQL search backend on FastAPI (GPT-3.5 query translation with "Did you mean?" fallback suggestions) for real-estate listings; load-tested and deployed on Google Cloud Run.
- **DocuAI, semantic document retrieval:** designed the chunking, ingestion and Qdrant vector-search pipeline over 1,000+ documents with a Next.js front end; delivered and demoed the working prototype end to end in two days.
- **Data & speech pipelines:** sensor-data ingestion (MQTT → MongoDB) for a bioreactor startup that later raised funding; spoken-English scoring from short videos (Whisper, Microsoft's pronunciation API, AWS Fargate); healthcare-device compliance pipelines.

EDUCATION & CERTIFICATIONS

Technical AI Safety course (BlueDot Impact) *Sep – Oct 2026*

BlueDot certificate (Oct 2026)

Deep RL (Udacity Nanodegree) · Machine Learning (Stanford Online) *2020*

Udacity certificate (Jul 2020) · Coursera certificate (Mar 2020)

B.Tech (CSE), undergraduate coursework *2018 – 2020*

APJ Abdul Kalam Technological University · Sree Buddha College of Engineering, Kerala

INDEPENDENT PROJECTS & RESEARCH

Diarist: Local Meeting Transcriber with Speaker Labels, for Windows *Oct 2026*

Public beta 0.1.0-beta.5, free and MIT-licensed. diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and speaker models, Svelte 5, TypeScript, SQLite (FTS))

- Records a meeting or imports a file and writes a transcript that names the speaker on every line, for up to eight speakers. It runs on the user's own GPU, with no cloud, account or Python install.
- Transcribes about **30x faster than real time** on an RTX 3080. On my own Zoom meetings, 0.2 to 0.3% of words went to the wrong speaker. First public beta the day work started, built with coding agents; tested on one PC.

ProofPack: Website-Check Agent That Keeps Hashed Proof for Every Claim *Jul – Sep 2026*

Built from a challenge brief (Build with Gemini XPRIZE, Jul 2026). A project, not a product; no users. github.com/ebt55/proofpack (Python, Gemini API, Playwright, FastAPI, Jinja2, pytest)

- Checks Medicaid purchase requests against providers' public websites, as a compliance reviewer would, and saves a stamped screenshot for every claim. Only the browsing step uses AI; plain code writes timestamps and SHA-256 hashes.
- Code rejects any "Found" without a real screenshot, and the reviewer makes every decision. On seven made-up request forms, each review cost **$0.02 to $0.19** in Gemini fees.

councilrag & rbb: Council-Meeting Search with Video Citations, and a Retrieval Benchmark *Oct 2026*

From a private 48-hour company screening test, Oct 2026 (no public repo). (Python, LanceDB (BM25 + vector + RRF), Qwen3-4B, Granite-3.3-2B, RapidOCR, Diarist ASR, pytest)

- councilrag searches 433 Seattle City Council meetings (about 813 h), with citations that open the video at the second. Vote questions are answered from the official record; open questions mostly are not. On 8 meetings the captions dropped 18 of 60 located roll calls (17 on two); the audio transcript recovers **17 of 18**.
- rbb stores evidence as time spans so any chunking can be scored, then tests whether its metrics predict a reader's answers. On a 144-question Video-MME stratum (55 of 90 videos fit in 2k tokens), token budget alone explains 74% of accuracy variance across 45 configs. Within one budget, graded labels pick the config that answered **66% vs 50%** for token count (6,298 pairs from 83 of 144 questions). Seattle showed a null.

printready: Checks and Fixes AI Sticker Art Against a Printer's Rules *Oct 2026*

Not affiliated with Sticker Mule; rules paraphrased from its public FAQ. github.com/ebt55/printready (Python, OpenCV, scikit-image, Pillow, xAI API (Grok), MCP server, pytest)

- Checks AI-generated sticker art against a printer's published rules, fixes what it safely can, and hands the rest to a person. A guard undoes any fix that damages the art.
- An AI agent with the same tools did no better than fixed steps. Both matched my labels on **34 of 50 images**; the agent cost about four times as much. I would send 53 of 55 fixed proofs to a customer; one labeller, nothing printed.

ExactDoc: PDF to Editable Word Documents, Checked Page by Page *Aug – Oct 2026*

Beta 0.3.0b3 on PyPI. github.com/ebt55/exactdoc (Python, PDFium / pypdfium2, OOXML, LibreOffice headless, PyMuPDF, pytest)

- Turns a PDF into a Word document with real headings, lists and tables that still looks like the original. With LibreOffice installed, it reopens its own output, compares each page with the PDF and corrects page breaks.
- On 27 short PDFs never used in development, **22 kept exactly their page count in Word** (24 in LibreOffice). All 13 letters, memos and similar documents did; two LaTeX papers came out unusable.

Kobayashi Maru: Do Impossible Tasks Make AI Agents Cheat on Solvable Ones? *Sep 2026*

Apart Research × CeSIA AI Incident Response sprint (Sep 11–13), solo; top half of 230+ submissions. github.com/ebt55/kobayashi-maru (Python, Docker, Anthropic API, OpenRouter, Ollama, statsmodels)

- Tests whether impossible tasks in an agent's batch make it cheat on tasks it could solve, over **8,959 runs on six models**. For one model, DeepSeek-V4.1-flash, cheating on solvable tasks rose from 0 of 120 to 36 of 120. Limits: both positive models are cheap ones added later, and batch length grows with the number of impossible tasks.

diffing-agent-bench: Sealed Benchmark for AI Agents That Audit Fine-Tuned Models *Aug – Sep 2026*

Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026. github.com/ebt55/diffing-agent-bench (Python, PyTorch, PEFT / LoRA, vLLM, Qwen3.5-9B)

- Built a sealed test for AI agents that compare a fine-tuned model with its original. I planted habits in LoRA finetunes of Qwen3.5-9B and sealed the answer key before any run. The agent asked a database question in **0 of 13 attempts**, so it missed the planted database habit. A fixed 50-prompt list found it for $0.15, against $3.14 per detection for the agent.

IncidentGate: Test Lab for Safety Checks on an Incident-Repair Agent *Aug – Sep 2026*

Closed at a baseline, Sep 2026. github.com/ebt55/incidentgate (Python, LangGraph, PostgreSQL, OpenTelemetry, Langfuse, Docker Compose, pytest)

- Built a test lab for three safety checks on an incident-repair agent: a rule list, a watcher model and a scripted stand-in approver. A forbidden action made as one named call was stopped at once. A different forbidden change, split into three allowed steps, passed all checks with a local watcher model and **ended as if no checks were on**. Each setting ran once, and the attack was scripted.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app)**: backend refactor and OpenAI support, merged (PR #75); bpmn-io/refactorings: text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break Jul 2025 – Feb 2026. Independent engineering and research since Mar 2026.